

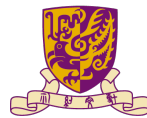
Network Programming

Computer Network Overview

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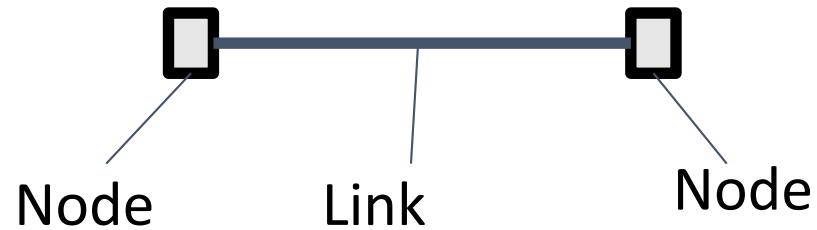


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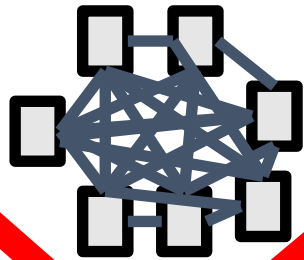
Basic building block – links



How to send data?

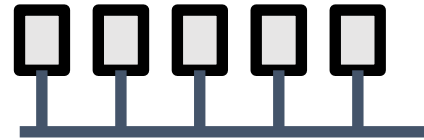
- When to talk – can either side talk at once?
- What to say – low-level format?

What if we want more nodes?



Wires for everybody!

Not scalable!

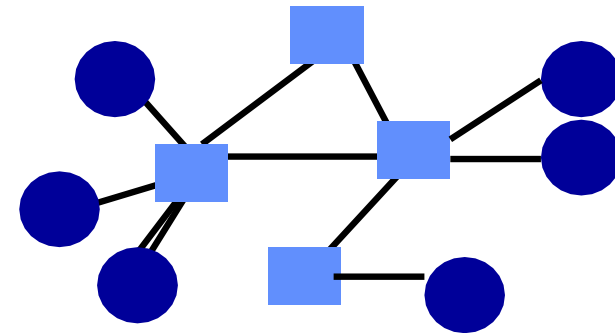


One wire

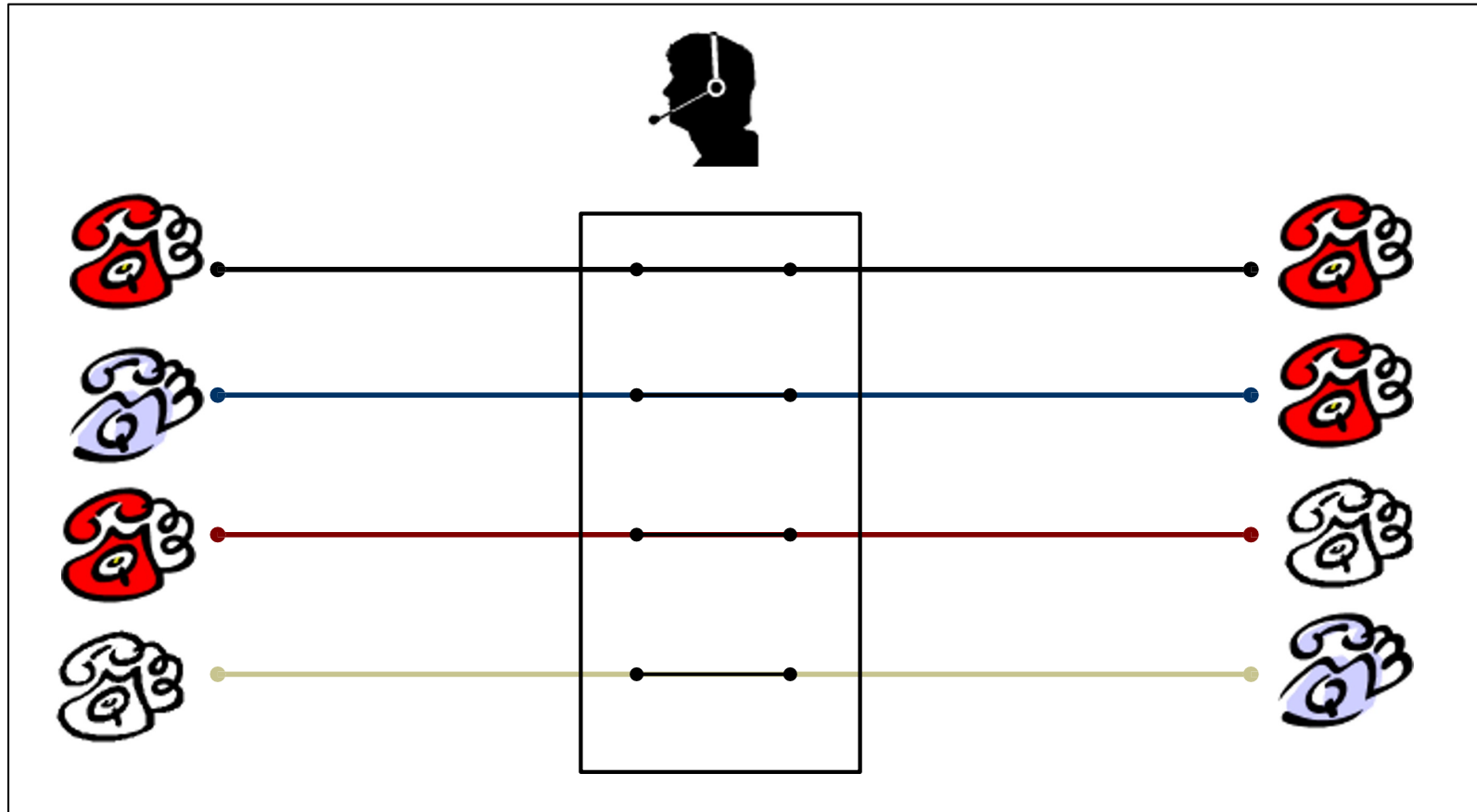
When to send?
Who should receive?

Multiplexing

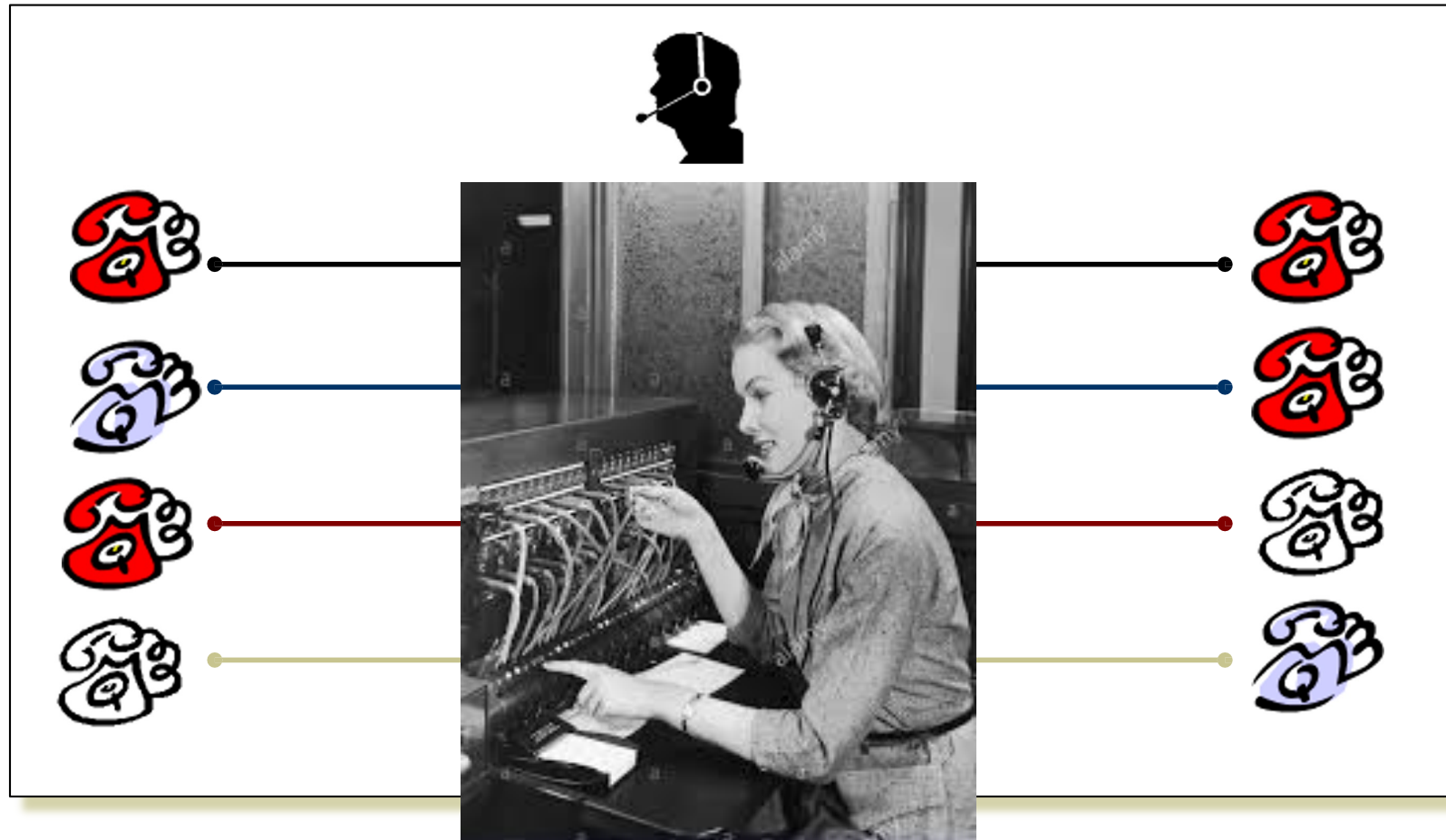
- Need to share network resources
- How? Switched network
 - Party “A” gets resources sometimes
 - Party “B” gets them sometimes
- Interior nodes act as “Switches”
- What mechanisms to share resources?



In the Old Days...Circuit Switching

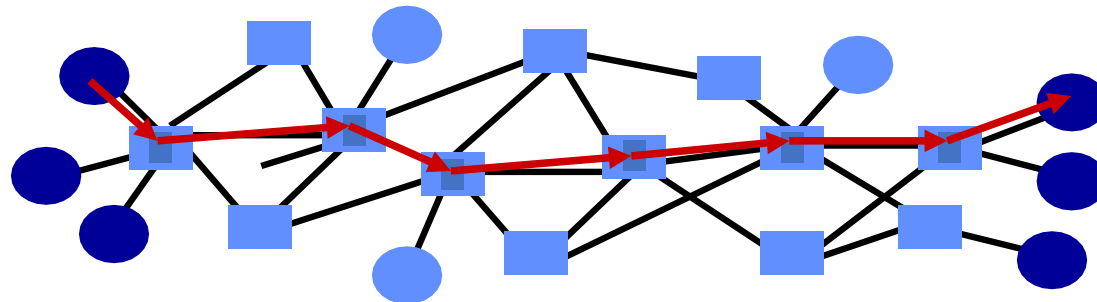


In the Old Days...Circuit Switching

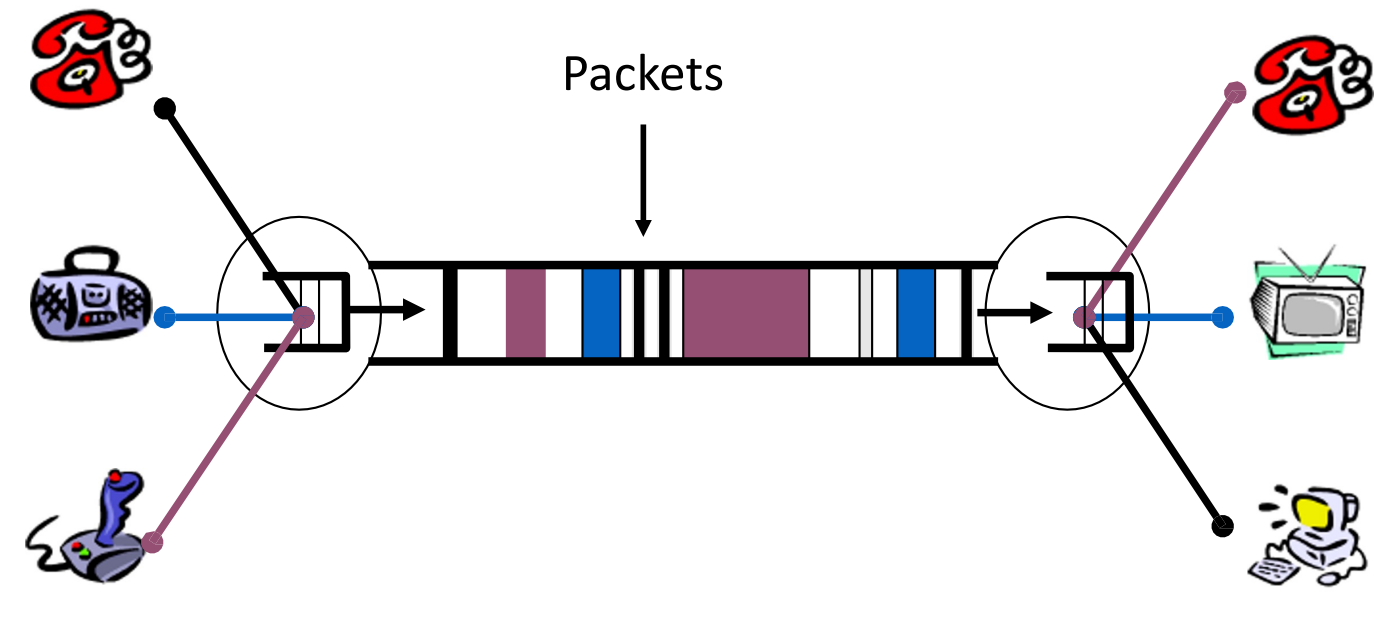


Packet switching

- Source sends information as self-contained packets
 - Have an address
 - Source may have to break up single message in multiple
- Each packet travels independently to the destination host
 - Switches use the address in the packet to determine how to forward it
 - Store and forward
- Analogy: a letter in surface mail



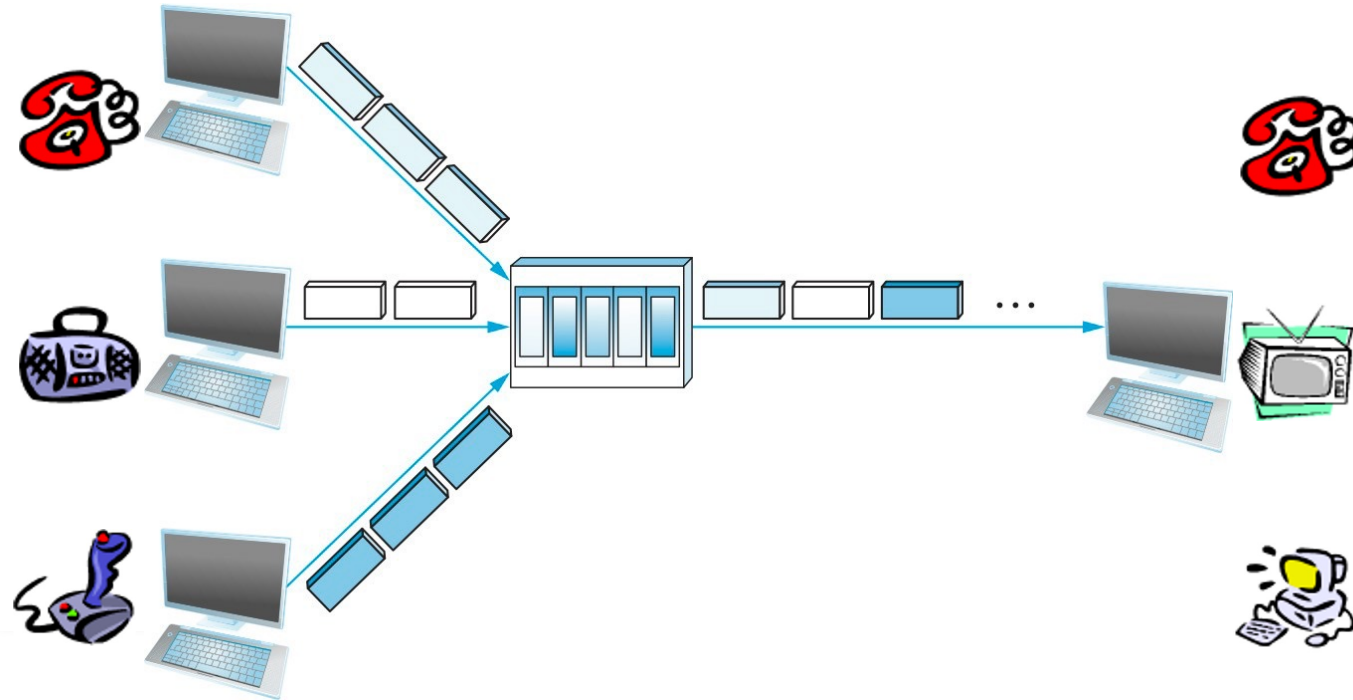
Packet switching – statistical multiplexing



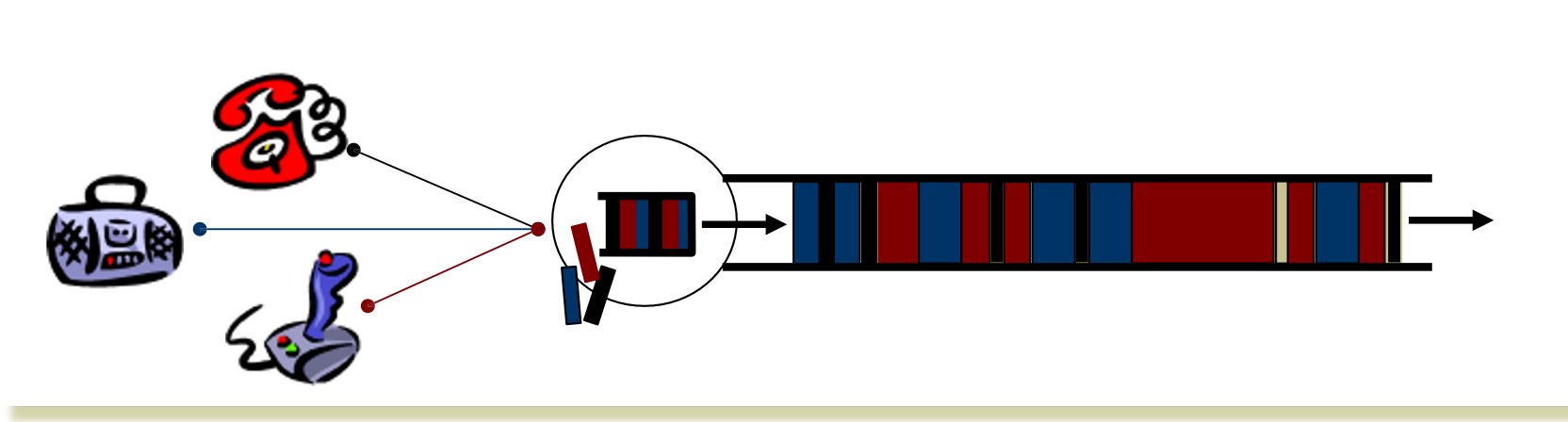
Switches arbitrate between inputs
Can send from **any input** that's ready

Efficiency!

Packet switching – statistical multiplexing



What if network is overloaded?



Solution: Buffering and Congestion Control

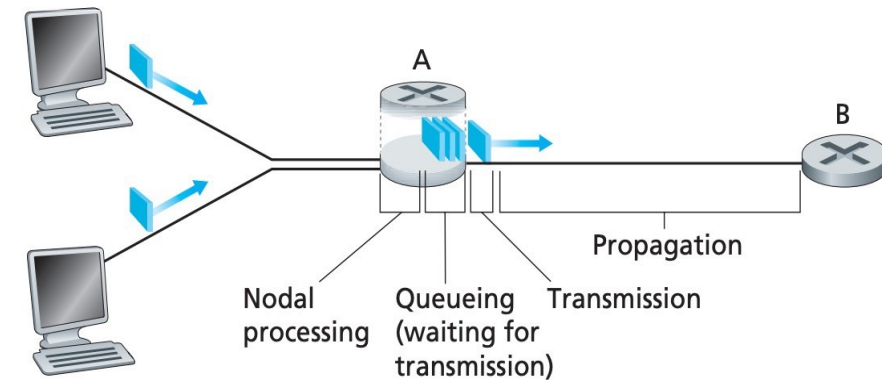
- Short bursts: buffer
- What if buffer overflows?
 - Packets dropped
 - Sender adjusts rate until load = resources → “congestion control”

Model of a communication channel

- **Latency:** how long does it take for the first bit to reach destination
- **Capacity:** how many bits/sec can we push through? (often termed “bandwidth”)
- **Jitter:** how much variation in latency?
- **Loss/Reliability:** can the channel drop packets?

Packet delay

- Propagation delay on each link
 - Proportional to the length of the link
- Transmission delay on each link
 - Proportional to the packet size
- Processing delay on each router
 - Depends on the speed of the router
- Queuing delay on each router
 - Depends on the traffic load and queue size



Packet delay

x_{mit} = transmission delay

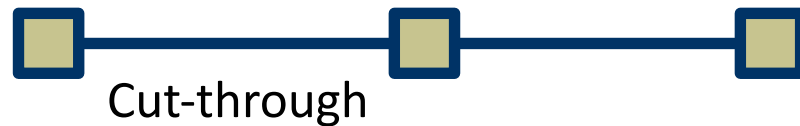
$Prop + x_{mit}$



$2 * (Prop + x_{mit})$

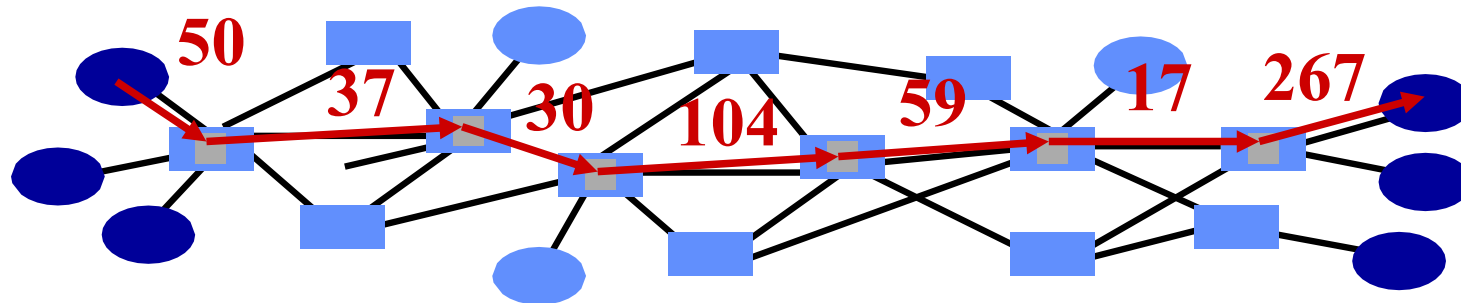


$2 * Prop + x_{mit}$



Sustained throughput

- When streaming packets, the network works like a pipeline
 - All links forward different packets in parallel
- Throughput is determined by the slowest stage
 - Called the bottleneck link
- Why some links are slow
 - Low link bandwidth
 - Many users sharing the link bandwidth

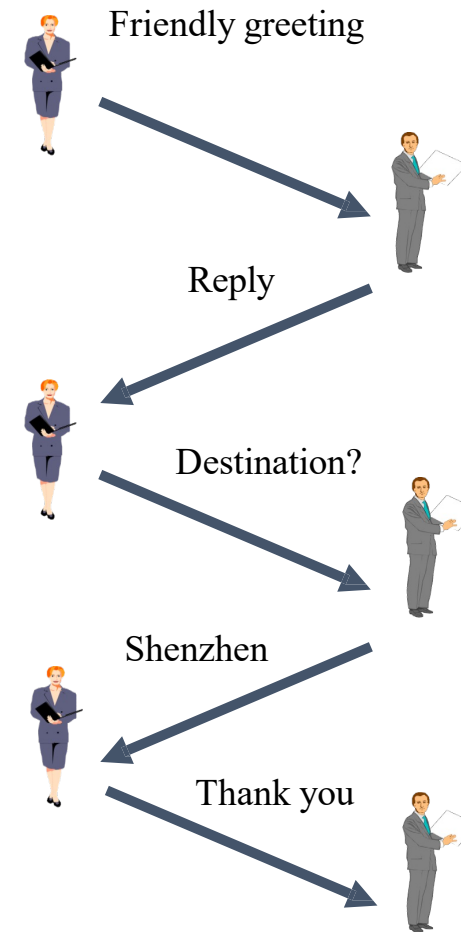


Can we know if the packet successfully arrives?
i.e., reliable transmission



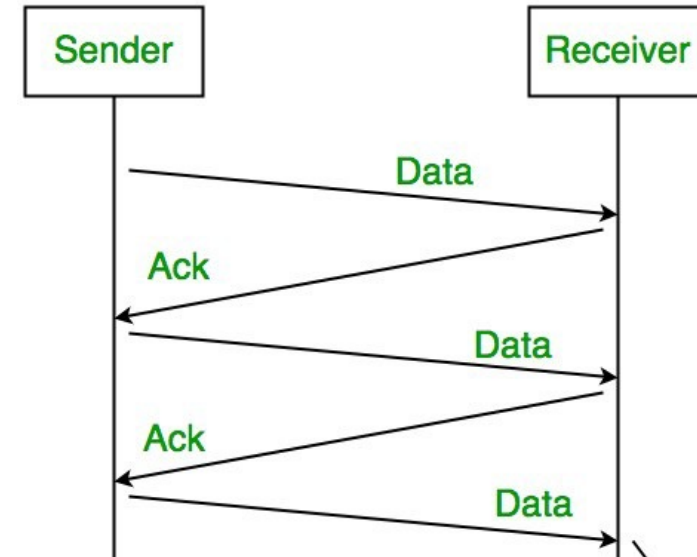
Network protocols

- Protocol
 - Agreement between parties on how communication should take place
- Stop & wait
 - Sender sends a single packet to receiver & waits for an acknowledgment
- Example
 - Polite conversation, buying airline ticket over phone



Network protocols

- A kind of reliable delivery
 - Acknowledgements (ACK) and timeouts
- Receipt of ACK
 - Sender knows the data was successfully delivered
- Timeouts
 - Retransmits original data if no ACK received after a reasonable timeout



Bandwidth calculation

Cross country latency of 1 bit

- Distance/speed = $5 * 10^6 \text{m} / 2 \times 10^8 \text{m/s} = 25 * 10^{-3} \text{s} = 25 \text{ms}$
- 50ms round-trip-time (RTT) for one bit

Link speed (capacity) 100Mbps

Packet size = 1250 bytes = 10 kbits

- Packet size on networks usually = 1500bytes across wide area or 9000bytes in local area

1 packet takes

- $10\text{k}/100\text{M} = .1 \text{ ms}$ to transmit
- 25ms to reach there
- ACKs are small → so 0ms to transmit
- 25ms to get back

Effective bandwidth = $10\text{kbits}/50.1\text{ms} = 200\text{kbits/sec}$ 😞

How should we send packets?

- What if we sent two packets before waiting for an ACK?
- What if we sent N packets?
- How many packets do we need to send before we use up the capacity of the link?

Answered in network protocols (next lectures)

When data gets Lost

Problem: Lost Data



GET index.html



Solution: Timeout and Retransmit



GET index.html



GET index.html



GET index.html



What if the data gets Corrupted?

Problem: Data Corruption



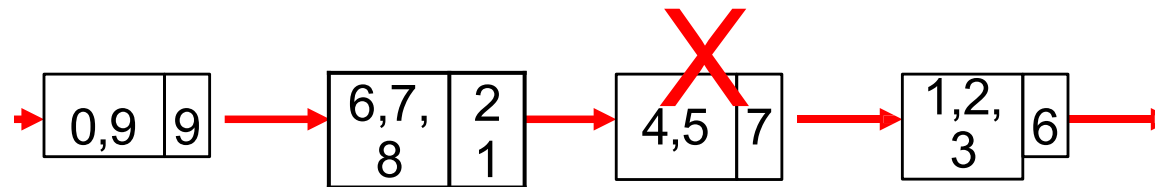
GET
index.html



GET
inrex.html

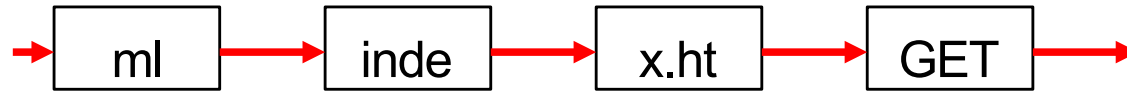


Solution: Add a *checksum*



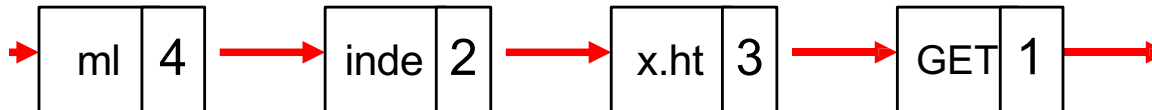
What if the data is Out of Order?

Problem: Out of Order



GET x.htindeml

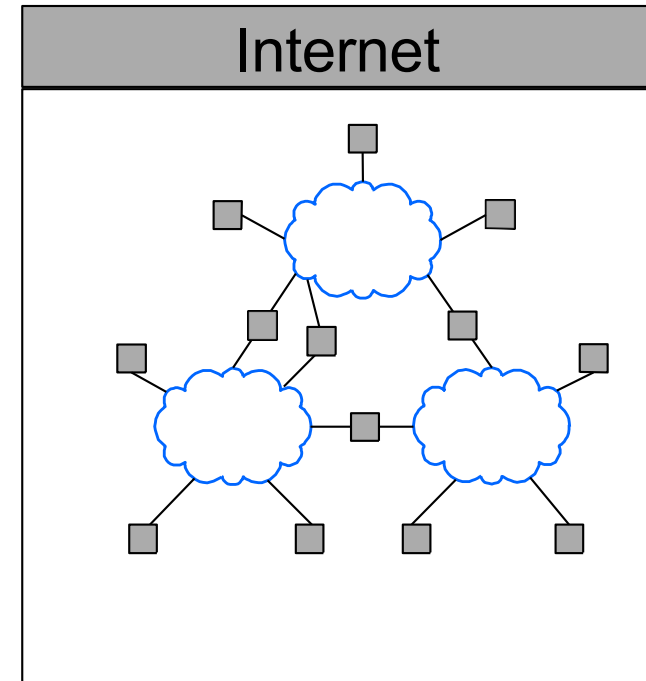
Solution: Add Sequence Numbers



GET index.html

Internet

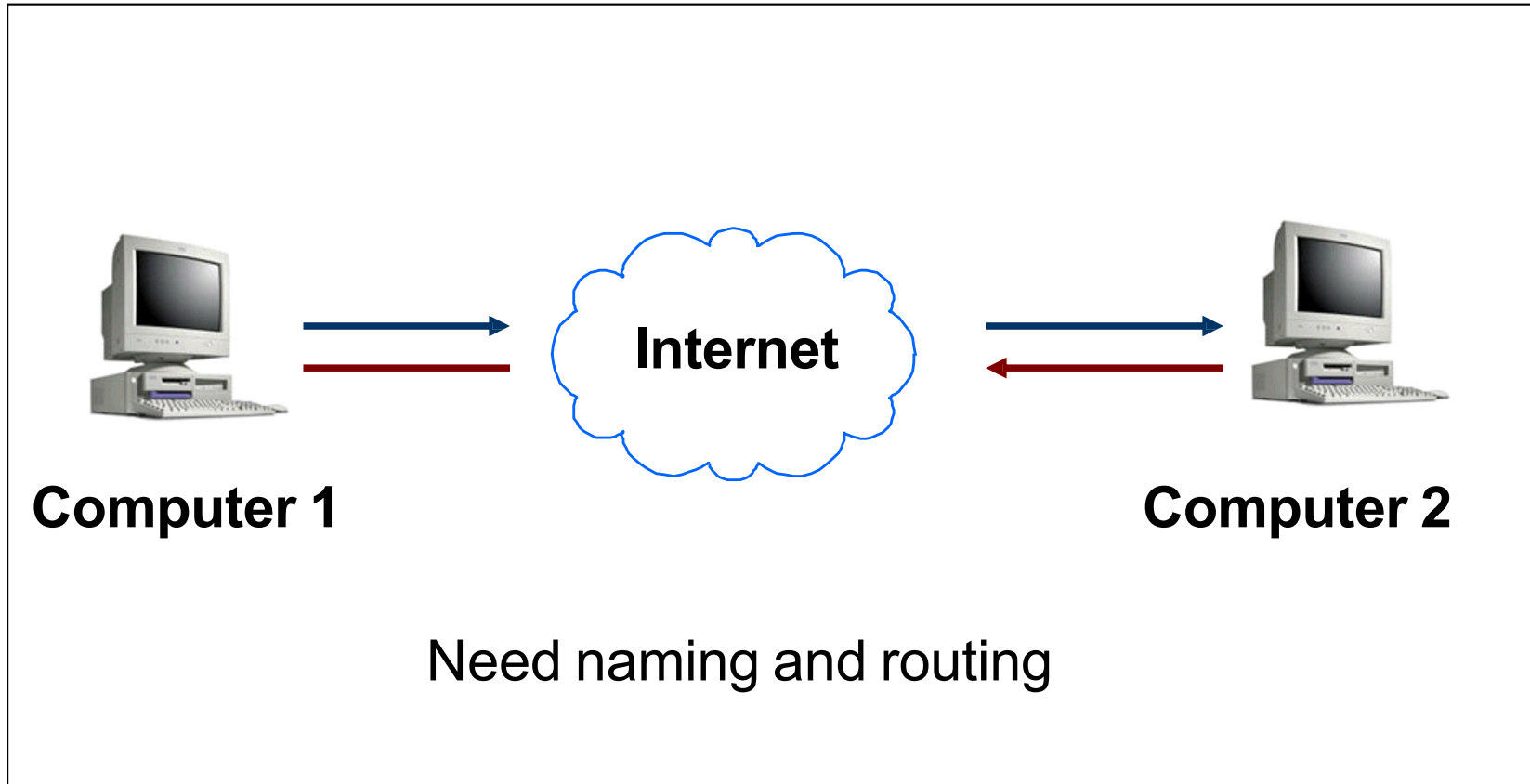
- An inter-net: a network of networks
 - Networks are connected using routers that support communication in a hierarchical fashion
 - Interconnected set of networks of the Internet Service Providers (ISPs)
 - > 40,000 different networks make up the Internet



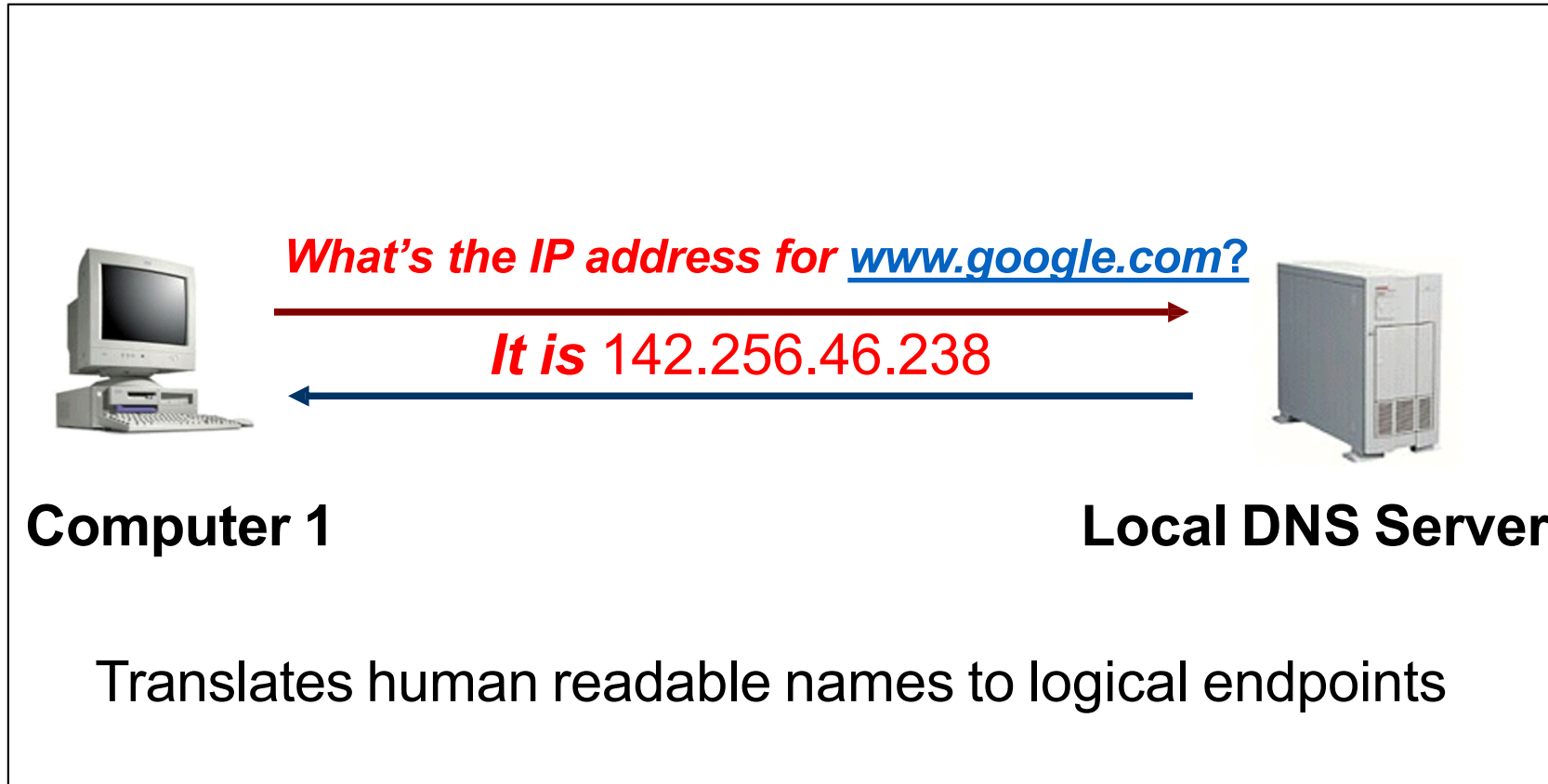
Other functions in addition to reliability?



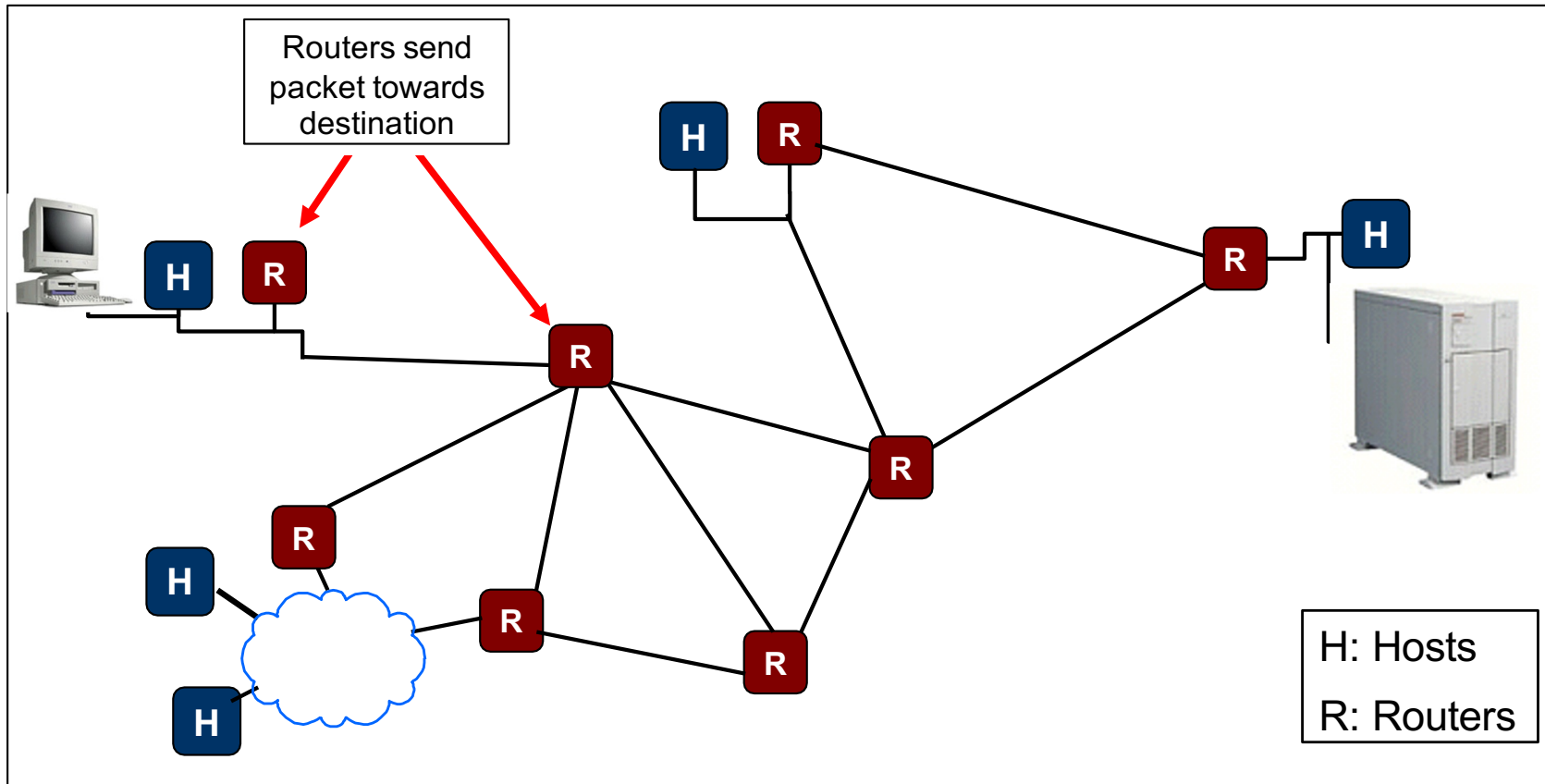
How to find nodes?



Naming



Routing



Networks with many functions

- Link
- Multiplexing
- Reliability
- Routing
- Addressing/naming (locating peers)

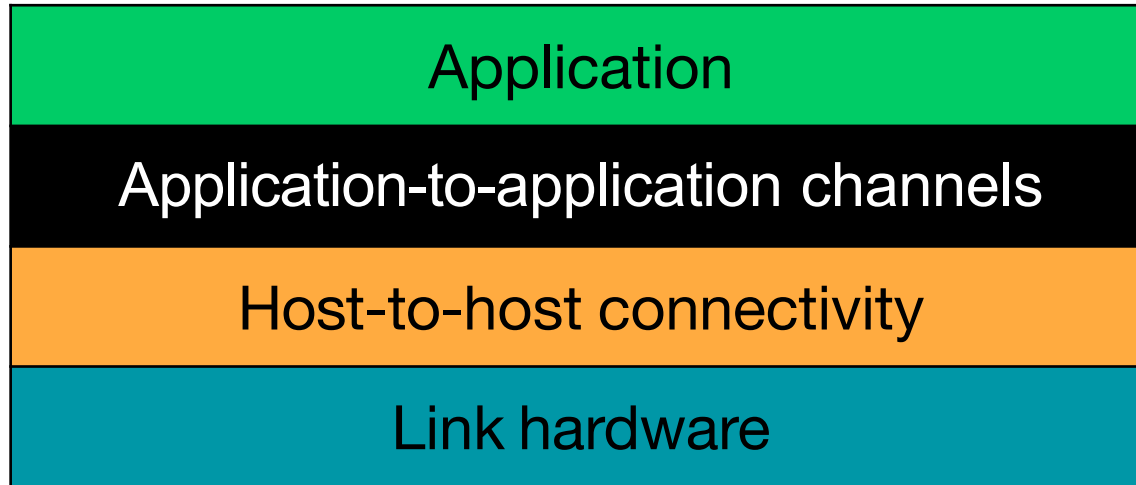
...

Where would you implement each function?

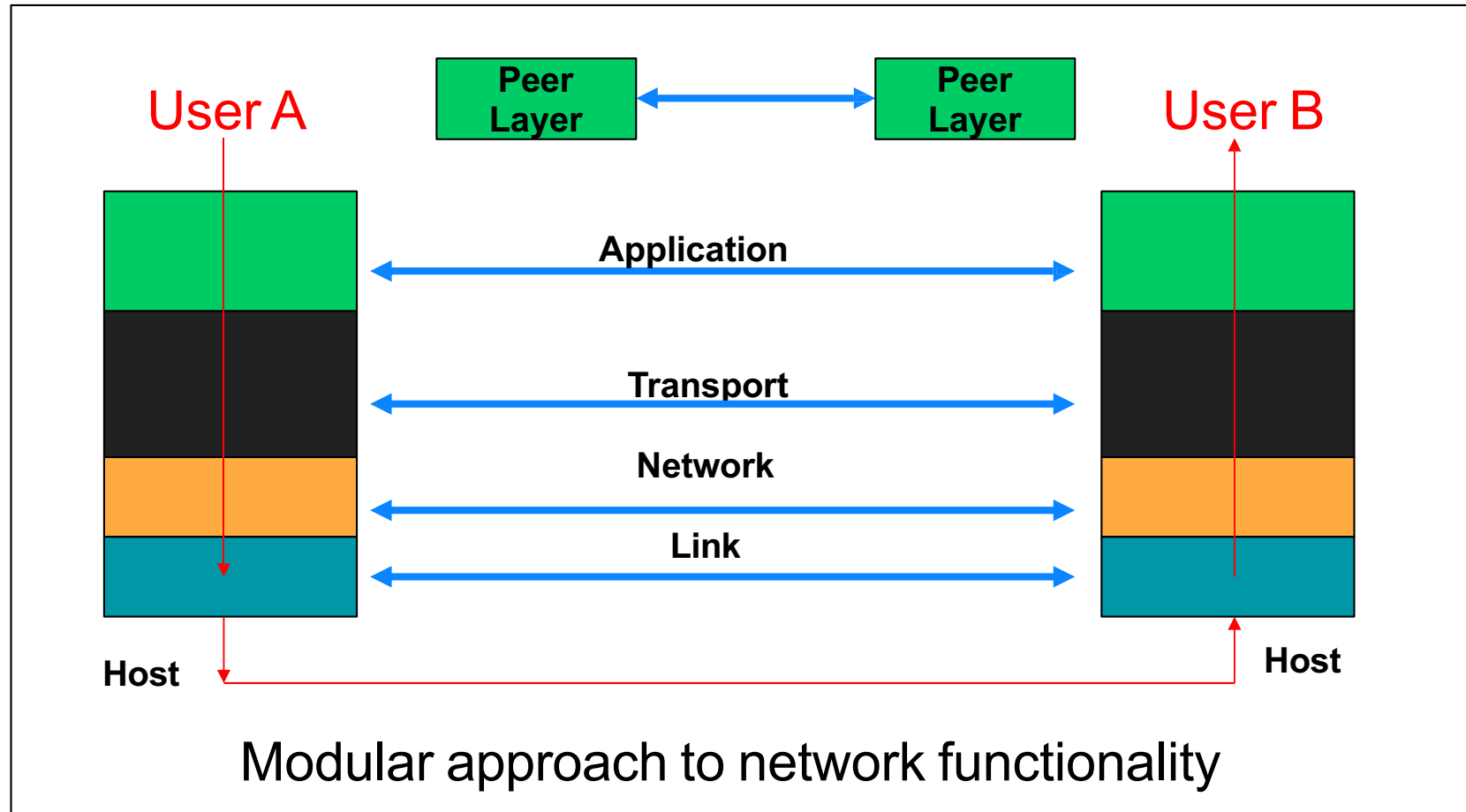
Layered design!

What is layering?

- Modular approach to network functionality



What is layering?



Layering characteristics

- Each layer relies on services from layer below and exports services to layer above
- Interface defines interaction with peer on other hosts
- **Protocols** define:
 - Interface to higher layers (API)
 - Interface to peer (syntax & semantics)
 - Actions taken on receipt of a messages
 - Format and order of messages
 - Error handling, termination, etc.
- Hides implementation - layers can change without disturbing other layers (black box)

Reference Model

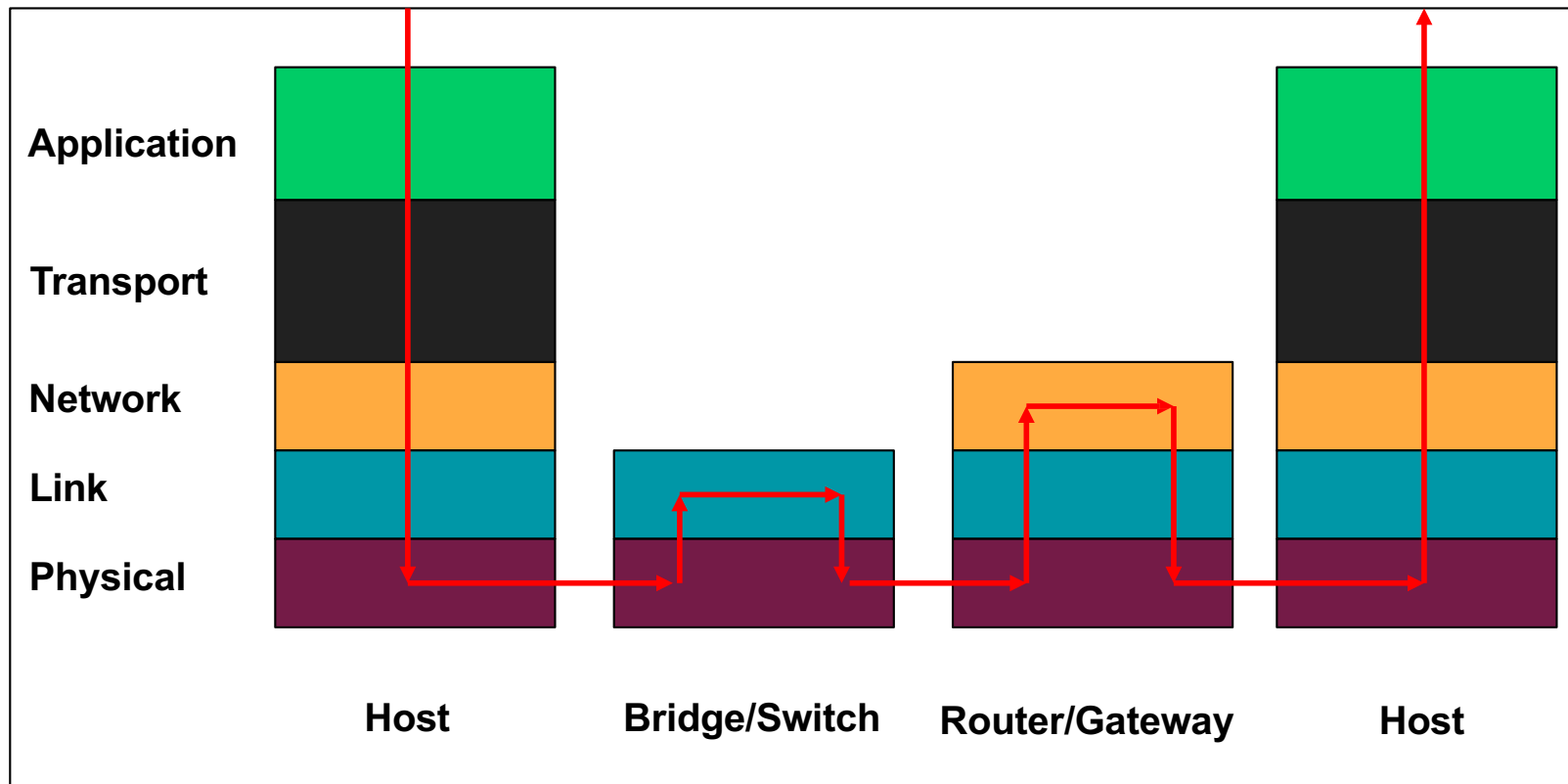
5	Application	Programs that use network service
4	Transport	Provides end-to-end data delivery
3	Network	Send packets over multiple networks
2	Link	Send frames over one or more links
1	Physical	Send bits using signals

Reference Model

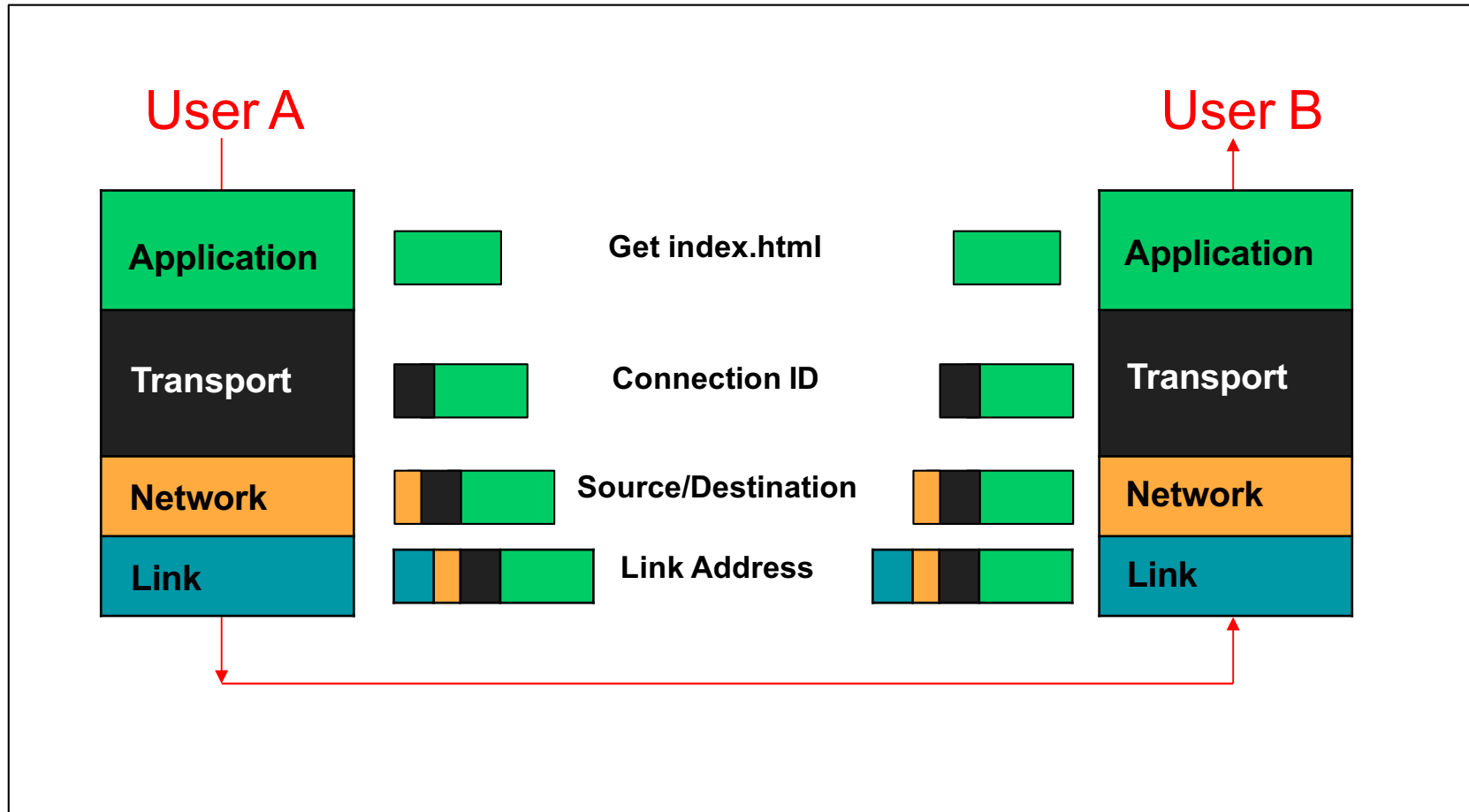
5	Application	HTTP, DNS, CDNs
4	Transport	TCP, UDP
3	Network	IP, NAT, BGP
2	Link	Ethernet, 802.11
1	Physical	wires, fiber, wireless

Internet protocol layering

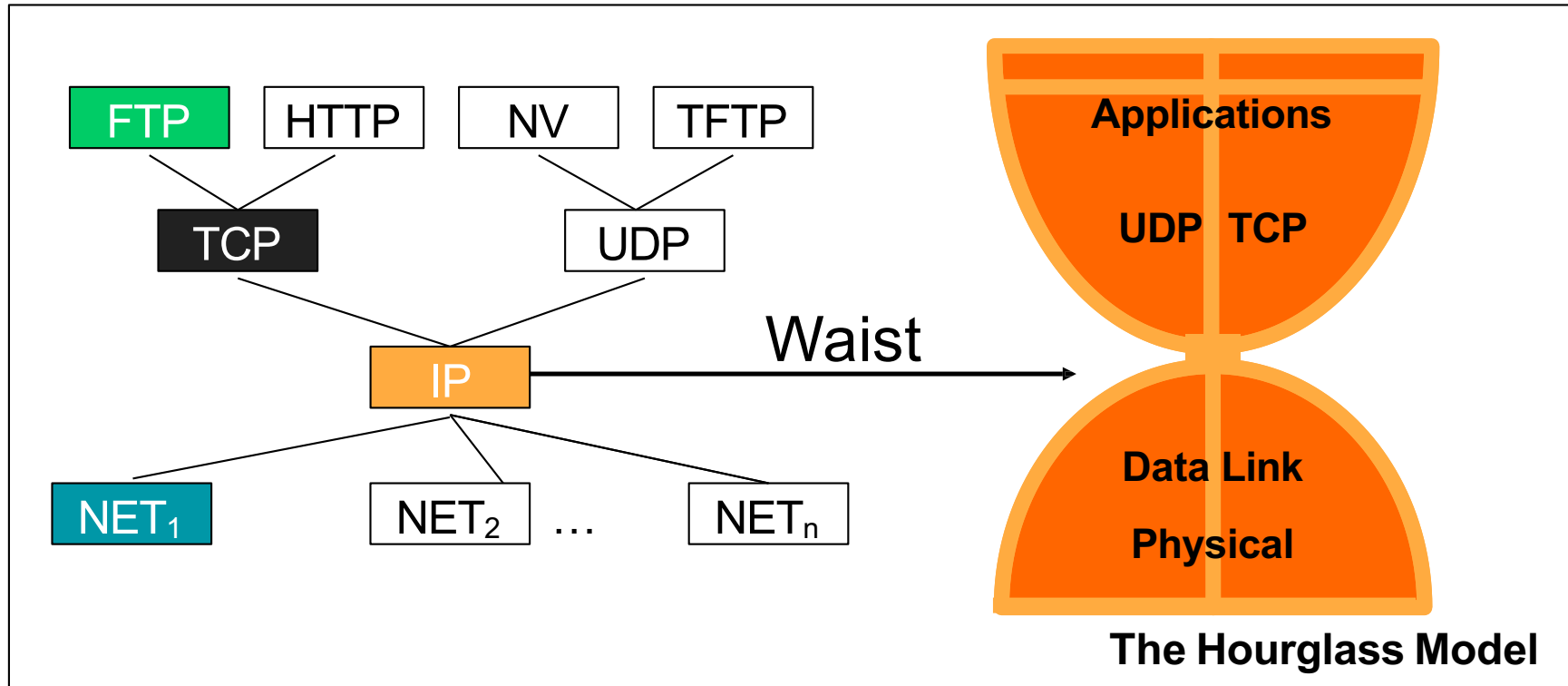
- Host-to-host communication over layers



Layer encapsulation



Internet protocol suite

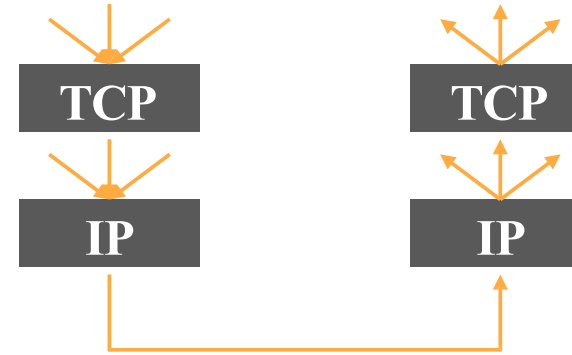
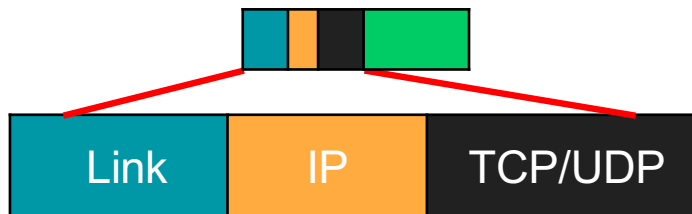


Interoperability: IP over anything, anything over IP

Any disadvantage of the “IP waist”?

Multiplexing and demultiplexing

- There may be multiple implementations of each layer
 - How does the receiver know what version of a layer to use?
- Each header includes a demultiplexing field that is used to identify the next layer



Example of IP header

V/HL	TOS	Length
ID		Flags/Offset
TTL	Prot.	H. Checksum
Source IP address		
Destination IP address		
Options..		

What networks do?

- Make and break connections
- Find a path through the network
- Transfers information reliably
- Transfers arbitrary length information
- Send as fast as the network allows
- Shares bandwidth among users
- Secures information in transit
- ...

Credit

- Some slides are adapted from course slides of 15-440 in CMU